Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Natural disasters severely disrupt transportation infrastructure, heavily complicating search and rescue (SAR) operations. Existing humanitarian logistics models often fail to capture these dynamic disruptions and the non-linear escalation of human suffering. To address this gap, we propose a two-stage stochastic Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). Our model simultaneously minimizes expected logistics costs – utilizing Daganzo's Continuous Approximation for scalable last-mile routing – and the expected maximum deprivation cost to enforce social equity. To solve this NP-hard problem, we develop a Priority-Based Non-dominated Sorting Genetic Algorithm II (PB-NSGA-II). Extensive computational experiments on standard benchmarks (AP, TR81) validate the algorithm's efficiency. Furthermore, a detailed case study of the flood-prone Vu Gia – Thu Bon river basin provides crucial managerial insights. The results highlight the necessity of dynamically adapting multi-modal network topologies to maintain resilience under extreme disaster scenarios.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [?, ?]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [?, ?].

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There were many great challenges in humanitarian operations during this period. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains [?], while key arteries such as National Highway 1 were severed by landslides, isolating communities for days [?]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster management involves four phases: mitigation, preparedness, response, and recovery [?]; Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [?, ?]. A comprehensive review of humanitarian facility location under uncertainty [?] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [?] document the growing need for models that jointly capture disruption, uncertainty, and equity. Critically, Holguín-Veras et al. [?, ?] argued that purely cost-based objectives neglect social welfare, and proposed *deprivation cost* – an exponential penalty on victim waiting time – as the appropriate humanitarian objective.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell's seminal formulations [?]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [?]. Sangsawang and Chanta [?] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [?].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [?, ?]. Under demand and supply uncertainty, robust formulations [?, ?] and prepositioning models [?] have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [?], a condition endemic to Central Vietnam's flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [?] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [?] applied a multi-objective evolutionary algorithm to multi-period dynamic emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [?] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. No existing work simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [?] is the closest precedent, yet it is deterministic and single-objective. The present paper closes this gap by proposing MO-IHLNDP – a two-stage stochastic multi-objective incomplete hub location model integrating Daganzo's CA for last-mile cost estimation – and PB-NSGA-II, a priority-based [?] evolutionary algorithm that efficiently reconstructs second-stage decisions across flood scenarios in Central Coastal Vietnam.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND) for flood disaster response in Central Vietnam, we frame the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP) – the capacitated, single-allocation variant of the HLP. The network comprises three main components: origins, hubs, and demands. Hubs refer to rescue stations of two kinds: the first is established in advance of the disaster, strategically positioned and pre-inventoried with amenities; the second is opened dynamically during response, prioritizing approachability to affected areas. **Demands** refer to victim groups discovered progressively during operations, while **origins** represent both government-planned forces and spontaneous volunteers.

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, urging the need for a dynamic solution [?]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [?]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that SAR operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

Finally, the model places strong emphasis on *social equity* by integrating the deprivation cost [?], and adopts Daganzo's continuous approximation [?] to obtain a tractable yet robust estimate of rescue time from each hub to its assigned demand points.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s: $c_{jks} = \min_{m \in \mathcal{M} \mid a_{jkms}=1} C_{jkm}$, $(+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$
\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned}
\tag{1}
$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [**?**]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost Θ_{kis} is pre-computed as a parameter:

$$\Theta_{kis} = \min_{m\in\mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is}\cdot\Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [**?**], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k\in\mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned}
&\text{Minimize} \quad Z_1, Z_2 \\
&\text{subject to} \quad \text{Constraints } (\textbf{??}) - (\textbf{??})
\end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{3}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{4}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{7}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \tag{12}$$

$$\begin{aligned}
\sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} \leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\
+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S}
\end{aligned} \tag{13}$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{15}$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \tag{16}$$

Constraints (**??**) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (**??**), while the reactive hubs are required to be placed within safe zones (**??**). Constraints (**??**)-(**??**) rigorously enforce the single-allocation property to active hubs via available links. Constraint (**??**) restricts flow to be trans-shipped on intact paths. Constraints (**??**) and (**??**) maintain flow conservation and capacity limits. Specifically, (**??**) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (**??**)-(**??**).

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [**?**]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [**?**] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency $(\lambda \cdot D)$ and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$; if none qualify, the safest hub is forced active. The proximity order π_k, ranking all hubs by ascending distance from anchor hub k, is pre-computed once.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score

$$\mathrm{Score}_i = w_0\, \widetilde{\mathrm{urgency}}_i + w_3\, \widetilde{\mathrm{isolation}}_i - w_1\, \widetilde{\mathrm{dist}}_i + \varepsilon_i, \tag{17}$$

where urgency $= \lambda_{is} D_{is}$, isolation $= 1/|\mathcal{H}_{\mathrm{act},i}|$, and dist $=$ minimum travel time to any active hub, each normalised to $[0,1]$; $\varepsilon_i \sim \mathcal{N}(0, \sigma)$ prevents premature convergence. Demands are processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i, the trial order follows π_{A_i}. Let $K = \max(1, \lceil w_5 \cdot |\mathcal{H}| \rceil)$. *Pass 1* selects the hub maximising $\mathrm{HubScore}(k) = w_1/\tau_{ikm^*} + w_2 \cdot \mathrm{residual}_k + w_4 \cdot \mathbf{1}[X_k = 1]$ among the K closest-to-anchor hubs that are active, reachable, and have positive residual inventory. If Pass 1 fails, *Pass 2* assigns i to the first active reachable hub from the remaining candidates (capacity violation permitted). If both passes fail, a *reactive fallback* opens the nearest safe inactive hub at cost F^a_{ks}; irrecoverable infeasibility accumulates as constraint violation.

Step 4 – Supply Routing and Objective Accumulation. Origins are greedily assigned to the most under-supplied active hub. Surplus inventory is redistributed via inter-hub trans-shipment with discount α. Z_1 and Z_2 are accumulated as probability-weighted expectations over all $s \in \mathcal{S}$. Algorithm **??** formalises Steps 1–4.

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s, instance data
Ensure: $(Z_1^s, Z_2^s, \mathrm{CV}^s)$
Activate $\mathcal{H}_{\mathrm{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \le \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\mathrm{act}} = \emptyset$ **then**
  Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score each demand i by Eq. (**??**); sort in descending order
for each demand i (in priority order) **do**
  $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
  Pass 1: $k^* \leftarrow \arg\max \mathrm{HubScore}(k)$ over first K active, reachable hubs with residual > 0
  if Pass 1 succeeds **then**
    Assign $i \rightarrow k^*$; deduct demand from residual inventory of k^*
  else if any active reachable hub remains **then**
    Pass 2: assign i to first such hub (capacity violation allowed)
  else
    Reactive fallback: open nearest safe inactive hub; add F_{ks}^a; $\mathrm{CV}^s \mathrel{+}= 1$
  end if
end for
Assign origins greedily; redistribute surplus via inter-hub trans-shipment (discount α)
Accumulate Z_1^s, Z_2^s
return $(Z_1^s, Z_2^s, \mathrm{CV}^s)$ =0

3.3 Genetic Operators and Selection

Selection uses binary tournament on Pareto rank with crowding distance as tiebreaker. *Crossover* ($p_c = 0.9$) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX ($\eta_c = 20$) to $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m = 1/|\mathbf{C}|$) applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation ($\eta_m = 20$) to $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle [**?**]: feasible solutions always dominate infeasible ones; among infeasible solutions, lower CV dominates. When a partial Pareto front must be split, tiebreaking proceeds by (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance in $\mathbf{X}$-space – preserving genotypic diversity in hub configuration and countering premature convergence.

4 Computational Experiments

4.1 Experimental Setup

Implementation. PB-NSGA is implemented in C++17. Data generation and result analysis are performed in Python 3. All experiments run on a standard workstation (Intel Core i7, 16 GB RAM).

Algorithm parameters. Population $N = 100$; generations $G = 200$; $p_c = 0.9$; $\eta_c = \eta_m = 20$; $p_m = 1/|\mathbf{C}|$ where $|\mathbf{C}| = 2|\mathcal{H}| + |\mathcal{I}| + 6$; decoder noise

$\sigma = 0.05$; Pass-1 window $K = \max(1, \lceil w_5 \cdot |\mathcal{H}| \rceil)$ (evolved per individual). The safety threshold is $\chi = 0.7$; discount $\alpha = 0.6$; $\gamma = 3.0$ kg/person; $\lambda_0 = 0.8$.

Performance metrics. Each algorithm is executed across 20 independent runs per instance; all indicators are reported as mean ± standard deviation, with statistical significance assessed via the Wilcoxon rank-sum test ($\alpha = 0.05$). We employ three complementary Pareto-front quality indicators:

- **Hypervolume (HV)** [?]: volume of objective space dominated by the approximation set, bounded by reference point $\mathbf{r} = (1.1, 1.1)$ in the $[0, 1]^2$ normalised space (nadir of the combined front). A larger HV indicates better convergence and spread.
- **IGD$^+$** [?]: average modified Hausdorff distance from each reference point to its closest solution in the approximation set. For small instances (AP10, AP20, AP25, AP40, CV-Small), the reference set is the *exact true Pareto front* obtained by complete enumeration (Section **??**); for larger instances, it is the combined non-dominated front pooled across all 20 runs and all algorithms. A smaller IGD$^+$ indicates closer proximity to the true front.
- **Set Coverage (C-metric)** [?]: $C(A, B)$ denotes the fraction of solutions in B dominated by at least one solution in A. We report $C(\text{PB-NSGA}, \text{NSGA-II})$ and $C(\text{NSGA-II}, \text{PB-NSGA})$; a high $C(A, B)$ with low $C(B, A)$ indicates algorithm A dominates a larger portion of B's Pareto front.

4.2 Dataset Description

Benchmark Instances (HLP → DRND Transformation) We adapt two widely-used hub location benchmarks into DRND format. The **AP** set [?] provides Euclidean instances of sizes $n \in \{10, 20, 25, 40, 50, 100\}$; the **TR81** dataset [?] is a real inter-city distance and flow matrix for 81 Turkish provincial capitals.

Each benchmark instance is transformed via the following procedure:

1. **Role assignment.** Nodes are ranked by a safety-flow score $s_i = f_i(1 - \bar{r} - 0.3\,\tilde{f}_i)$, where f_i is total incident flow and $\tilde{f}_i$ its normalised value. Top-p nodes ($p = \lfloor 0.3n \rfloor$) are designated hubs; the remaining high-outflow nodes become origins/suppliers.
2. **Scenario generation.** Three flood scenarios (mild, severe, extreme) are assigned probabilities $(\pi_1, \pi_2, \pi_3) = (0.60, 0.30, 0.10)$. For each scenario, a random epicenter is drawn; road disruption penalties decay exponentially with distance ($\sigma \in [0.25, 0.35] \times d_{\max}$).
3. **Calibration.** Distances are normalised to a 500 km scale. Supply is set to 1.5–2.5× total demand. Hub fixed costs are proportional to average road distance (km-units). Three transport modes operate concurrently (Table **??**).

Central Vietnam Case Study Instances Two instances are constructed to model flood relief operations across the **Vu Gia – Thu Bn – Perfume** river basin, spanning four disaster-prone provinces: à Nng, Qung Nam, Tha Thiên-Hu, and Qung Ngãi.

Table 1. Transport mode parameters used for all instances.

Mode	Vehicle	Speed (km/h)	Cost ($/km)	Capacity (pers.)	Disruption
0	Truck	35	2.0	60	Road-dependent
1	Motorboat	25	5.0	25	Low (flood-resilient)
2	Helicopter	150	40.0	10	None (always available)

- **CV-Small**: $|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$ – district-level planning instance.
- **CV-Large**: $|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$ – commune-level regional instance.

Node georeferencing. Demand nodes are georeferenced to real administrative units (districts or communes) from four provinces. Base population follows a coastal-proximity model, $P_i \approx 500 + 6{,}500 \cdot f_{\text{coast}}$, reflecting the densely-settled coastal plains of Central Vietnam. Candidate hub locations are selected from logistics facilities, rescue stations, and military depots with confirmed multi-modal access. Supply origins are drawn from seaports (à Nng, Dung Qut, Chu Lai, Sa Kỳ), river ports (Thun An), and highland border logistics nodes.

Multi-criteria auxiliary risk. Each node u receives a static intrinsic flood-risk score $r_u^a \in [0, 1]$, computed as the weighted sum of four geographic criteria following [**?**]:

$$r_u^a = 0.25\, C_1(u) + 0.35\, C_2(u) + 0.20\, C_3(u) + 0.20\, C_4(u), \tag{18}$$

where C_1 is a topographic inundation index (decreasing with elevation, proxied by longitude relative to the highland–coastal gradient); C_2 is hydrological proximity to five river systems (Vu Gia, Thu Bn, Perfume, Trng Giang, Trà Bng) modelled via Gaussian decay ($\sigma = 18$ km); C_3 is coastal storm-surge exposure ($\sigma = 40$ km); and C_4 is river-delta accumulation risk centred on five historically inundated deltaic zones.

Risk intervals and scenario generation. Rather than assigning a single risk value per node, each node is assigned a risk interval $[r_u^{\min}, r_u^{\max}]$ whose width and ceiling scale monotonically with r_u^a:

$$r_u^{\min} = 0.05 + 0.20\, r_u^a, \qquad r_u^{\max} = r_u^a + 0.15(1 - r_u^a).$$

A safe inland node ($r_u^a = 0.10$) thus spans [0.07, 0.22], while a highly exposed delta node ($r_u^a = 0.80$) spans [0.21, 0.83]. Three scenarios (mild, severe, extreme) are generated with probabilities (0.60, 0.30, 0.10). Each scenario samples $n_s \in \{1, 2, 3\}$ flood epicenters from demand nodes with probability proportional to r_u^a; scenario risk for node u is then drawn from its interval, modulated by Gaussian exposure decay ($\sigma_e = 85$ km) from the epicenters. Road link (u, v) is disrupted with probability $\min(0.97,\ \beta_s \cdot \bar{r}_{uv})$, where $\beta_s \in \{0.25, 0.55, 0.88\}$ for mild, severe, and extreme events; water and air modes remain available under all scenarios.

Transport and capacity calibration. Road travel times combine Haversine distance with a fixed tortuosity factor (1.35, reflecting Central Vietnam's mountain roads) and a terrain multiplier ranging from 1.0 at the coast to 1.8 in the deep highlands. Demand volumes are scenario-specific: $D_{is} = P_i \cdot (0.05 + 0.85\, r_{is}) \cdot \mu_s$, where severity multipliers $\mu_s \in \{1.0, 1.8, 2.8\}$ capture the escalating relief burden. Last-mile distribution cost is modelled by the Daganzo continuum-approximation formula [**?**] with scenario-specific circuity $\phi \in \{0.57, 0.70, 0.85\}$ for the three scenarios. Hub capacities are calibrated post-generation so that aggregate installed capacity is 3–6× the worst-case total demand, with each hub's share scaled by its terrain factor to account for the larger staging areas feasible in flatter coastal zones.

4.3 Experiment 1: Algorithm Efficiency on HLP Benchmarks

Comparison Algorithms We compare **PB-NSGA** against the standard **NSGA-II** [**?**] as a direct baseline. NSGA-II uses the same population size ($N = 100$), generations ($G = 200$), and genetic operators but replaces the priority-based decoder with a direct full variable encoding (all second-stage routing variables). This comparison isolates the contribution of the priority-based decoding scheme. Both algorithms are seeded identically across 20 runs.

Ground-truth fronts via complete enumeration. To provide an objective quality reference for small benchmark instances, we apply a *complete enumeration* procedure that exhausts all $2^{|\mathcal{H}|}$ hub opening configurations. For each configuration $\mathbf{X}$, the second-stage sub-problem is solved via a structured heuristic (multiple candidate $\mathbf{R}$, $\mathbf{A}$, $\mathbf{W}$ settings) to approximate the true objective values, and only non-dominated solutions are retained. A 5-minute time limit is imposed per instance. This yields an exact (or near-exact) true Pareto front for AP10, AP20, AP25, AP40, and CV-Small ($|\mathcal{H}| \leq 10$), which serves as the reference set for IGD$^+$ computation on these instances.

Results Table **??** reports HV, IGD$^+$, Pareto front size, and CPU time for the AP and TR81 benchmark instances. Values are mean ± standard deviation over 20 runs.

Discussion. For small instances (AP10–AP40, CV-Small), IGD$^+$ is measured against the exact true Pareto front from complete enumeration, providing an absolute quality reference independent of the competing algorithm. For AP50, AP100, and TR81 − where exhaustive enumeration is infeasible − the combined non-dominated front across all 20 runs and both algorithms serves as the reference. We also report C-metric pairwise between PB-NSGA and NSGA-II; $C(A, B) \gg C(B, A)$ indicates that A's front dominates a substantially larger fraction of B's solutions. The anchor-hub mechanism ($\mathbf{A}$ chromosome) is expected to be particularly beneficial on larger instances, where the full variable encoding of standard NSGA-II disperses search effort across an exponentially larger space.

Table 2. Experiment 1: Algorithm efficiency on HLP benchmark instances (mean ± std, 20 runs). Best mean per instance in **bold**. C-metric: $C(\text{PB-NSGA}, \text{NSGA-II})$ / $C(\text{NSGA-II}, \text{PB-NSGA})$.

Instance	Algorithm	HV (norm.)	IGD$^+$	#Pareto	Time (s)	C-metric
AP10	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
AP20	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
AP25	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
AP40	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
AP50	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
AP100	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	
TR81	PB-NSGA	**–**	**–**	–	–	– / –
	NSGA-II	–	–	–	–	

4.4 Sensitivity Analysis

To assess how key operational parameters affect solution quality, we conduct one-at-a-time sensitivity experiments on the CV-Small instance using PB-NSGA with 10 independent runs per configuration, varying one parameter at a time while holding others at their baseline values ($\chi = 0.7$, $\pi_3 = 0.10$, transport costs at baseline).

Safety threshold χ. We vary $\chi \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. A stricter (lower) threshold restricts the set of safe hubs under disruption, forcing more reactive openings and increasing Z_1; a lenient (higher) threshold allows more hubs to remain operational at higher risk, elevating Z_2.

Scenario probability weight π_3. We test $\pi_3 \in \{0.05, 0.10, 0.15, 0.20\}$ (with π_1 and π_2 adjusted proportionally), quantifying how the optimal pre-positioning strategy shifts as planners assign greater probability to worst-case extreme-flood events.

Transportation cost multiplier β. All per-km costs are scaled by $\beta \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$, testing the modal shift from road to water/air transport as surface-route costs increase.

4.5 Experiment 2: Case Study – Central Vietnam Flood Relief

Comparison Algorithms For the Vietnam case study we include two additional multi-objective solvers alongside PB-NSGA:

– **NSGA-II** [?]: standard baseline (same encoding as Experiment 1).

Table 3. Sensitivity of mean HV to key operational parameters (CV-Small, 10 runs each; baseline values in parentheses).

Parameter	Level	HV (mean)
Safety threshold χ	0.5	–
	0.6	–
	0.7 *(baseline)*	–
	0.8	–
	0.9	–
Extreme probability π_3	0.05	–
	0.10 *(baseline)*	–
	0.15	–
	0.20	–
Cost multiplier β	0.5	–
	0.75	–
	1.0 *(baseline)*	–
	1.5	–
	2.0	–

– **NSMA** [**?**]: Non-dominated Sorting Memetic Algorithm, a state-of-the-art hybrid incorporating local search phases.

All three algorithms use identical budget ($N = 100$, $G = 200$) and are evaluated over 20 seeds per instance.

Results Figure **??** shows the combined Pareto fronts (pooled across 20 seeds) for CV-Small and CV-Large. Table **??** summarises HV, IGD$^+$, and Pareto front size.

Table 4. Experiment 2: Case study results – Central Vietnam instances (mean ± std, 20 runs). Best mean per instance in **bold**. C-metric column: C(PB-NSGA, Competitor) / C(Competitor, PB-NSGA).

Instance	Algorithm	HV (norm.)	IGD$^+$	#Pareto	C-metric
CV-Small	PB-NSGA	**–**	**–**	–	–
	NSGA-II	–	–	–	– / –
	NSMA	–	–	–	– / –
CV-Large	PB-NSGA	**–**	**–**	–	–
	NSGA-II	–	–	–	– / –
	NSMA	–	–	–	– / –

Fig. 1. Combined Pareto fronts for CV-Small (left) and CV-Large (right). Each front pools all non-dominated solutions across 20 seeds per algorithm.

Managerial Insights The Pareto front of CV-Small reveals a convex cost-deprivation trade-off. The shape reflects the non-linear structure of the deprivation objective: reducing Z_2 in the high-cost region of the frontier requires disproportionate investment in additional hub capacity and air-/water-mode transport. Humanitarian agencies with limited budgets should identify their operating point on this frontier rather than minimising cost alone.

Network topology analysis highlights scenario-driven structural shifts. Under the *mild* scenario, road transport from a moderate set of candidate hubs provides adequate coverage. Under the *extreme* scenario, a subset of hubs exceeds the safety threshold ($r_{ks} > \chi$) and becomes unavailable, triggering automatic modal substitution toward waterway and helicopter routes. The remaining active hubs absorb greater pre-positioned inventory, partially compensated by inter-hub trans-shipment. This dynamic adaptation confirms the operational value of the two-stage stochastic formulation: a deterministic model calibrated to a single representative scenario would systematically under-invest in resilient transport modes and backup hub capacity.

For CV-Large, the anchor-hub mechanism encoded in $\mathbf{A}$ proves particularly valuable. Commune-level demand nodes form distinct geographic clusters aligned with river basins and coastal plains; the $\mathbf{A}$ chromosome implicitly partitions the search space into hub catchment areas, enabling PB-NSGA to maintain diversity across 20 hub candidates while allocating 100 demand nodes efficiently.

5 Conclusion and Future Work

The Central Vietnam case study, constructed from real geographic and demographic data across four flood-prone provinces, illustrates the practical value of the framework. The two-stage stochastic model responds to infrastructure disruption by dynamically shifting to water and air transport and redistributing inventory, demonstrating that deterministic planning would systematically under-invest in network resilience.

Limitations. The model assumes a fixed discrete scenario set; continuous risk distributions are not yet supported. The decoding heuristic is greedy and may miss globally optimal demand assignments for highly constrained instances.

Future work will extend the approach to rolling-horizon multi-period relief operations and incorporate real-time sensor feeds for online re-optimization. The complete enumeration baseline naturally motivates exact solver development (branch-and-cut or Benders decomposition) for medium-scale instances to tighten IGD$^+$ reference fronts and strengthen convergence guarantees.

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